

Corrective Action Report —

CAR-2025-008

Field	Value
Document ID	CAR-2025-008
Issue Date	2025-10-28
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CAR Status	Open
Subject	Pinnacle Energy Maintenance Service Documentation References Deprecated Pre-Acquisition Procedures; Audit of Work Performed at SPP-WIND-025
Affected Region / Site / Asset	SPP / SPP-WIND-025 / WIN-10007, WIN-10132 (Nordex assets)
Severity	Medium
Owner	M. Beasley, Vendor Manager — Field Services; co-owner G. Calderon, OEM Liaison
Original Target Close Date	2026-02-28
Distribution	Procurement; Vendor Management Office; SPP Site Manager; Engineering Standards Committee; Documents Control

1. Finding

Pinnacle Energy Maintenance ("Pinnacle") performs scheduled and condition-based maintenance for SPP-WIND-025 under Vendor Repair Agreement **VRA-2023-018**. Pinnacle's field-service documentation references service procedures from **NorthStar Wind Services** — a company Pinnacle acquired in 2022. NorthStar Wind Services' procedure library was last revised in 2021 and has not been brought current with subsequent OEM bulletins, Cascadia procedural standards, or applicable IEC/IEEE consensus updates.

A focused audit of 18 work orders executed by Pinnacle at SPP-WIND-025 between 2025-01-01 and 2025-09-30 identified procedural-currency deficiencies on 11 of 18

work orders (61%). The deficiencies range in severity from cosmetic (vendor procedure cites a part-number revision that was superseded but the part installed is equivalent) to material (vendor procedure omits a torque-verification step added by Nordex Service Bulletin NDX-SB-2024-006 affecting blade-root bolt assemblies on the affected fleet).

2. Evidence

2.1 Audit sample summary

Work Order	Date	Asset	Pinnacle Procedure Cited	Currency Issue
WO-2025-01147	2025-02-09	WIN-10007	NSW-WPM-04 Rev. 2 (2021-08)	Cites pre-acquisition vendor name; superseded by NDX-SVC-04 Rev. 5
WO-2025-01206	2025-03-04	WIN-10132	NSW-WPM-12 Rev. 1 (2020-11)	Missing blade-root torque step per NDX-SB-2024-006
WO-2025-01310	2025-04-15	WIN-10007	NSW-WPM-04 Rev. 2 (2021-08)	Same as 01147; recurring
WO-2025-01355	2025-04-29	WIN-10132	NSW-WPM-07 Rev. 3 (2021-04)	Cites obsolete oil grade (Mobil SHC 632 – superseded by SHC XMP 320)
WO-2025-01402	2025-05-18	WIN-10007	NSW-WPM-12 Rev. 1 (2020-11)	Missing NDX-SB-2024-006 step
WO-2025-01451	2025-06-03	WIN-10132	NSW-WPM-09 Rev. 2 (2021-06)	Gearbox endoscopy procedure references discontinued borescope model
WO-2025-01498	2025-06-22	WIN-10007	NSW-WPM-04 Rev. 2 (2021-08)	Recurring

Work Order	Date	Asset	Pinnacle Procedure Cited	Currency Issue
WO-2025-01567	2025-07-29	WIN-10132	NSW-WPM-12 Rev. 1 (2020-11)	Missing NDX-SB-2024-006 step
WO-2025-01612	2025-08-19	WIN-10007	NSW-WPM-04 Rev. 2 (2021-08)	Recurring
WO-2025-01660	2025-09-11	WIN-10132	NSW-WPM-12 Rev. 1 (2020-11)	Missing NDX-SB-2024-006 step
WO-2025-01701	2025-09-26	WIN-10007	NSW-WPM-07 Rev. 3 (2021-04)	Obsolete oil grade

The remaining 7 work orders cited current Nordex procedures correctly.

2.2 Service Bulletin NDX-SB-2024-006

Nordex SE issued NDX-SB-2024-006 ("Blade-Root Bolt Audit — Field Bulletin") on 2024-04-18 introducing a torque-verification step at the M48 root-bolt circle for the deployed turbine model. The bulletin was distributed to all known operators of the affected fleet, including Cascadia. Cascadia Engineering received and acknowledged the bulletin (Documents Control receipt 2024-04-22). The bulletin was forwarded to Pinnacle on 2024-04-26 via Cascadia Vendor Management.

Pinnacle has not updated their internal procedure library to reflect NDX-SB-2024-006. The audited work orders confirm execution against the pre-2024 procedure.

2.3 Linked outage

- WIN-10007 (Nordex, SPP-WIND-O25) has 12 forced outages in FY2025 with 13,538 MWh lost. While no single failure has been attributed to blade-root bolt issues to date, the bolt-torque step is a precursor inspection whose absence elevates the risk of an undetected fatigue-crack initiation. The risk is latent rather than realized.
- WIN-10132 (Nordex, SPP-WIND-O25) has 10 forced outages and 13,173 MWh lost.

2.4 Cost and labor

Pinnacle bills at **\$245/hr** — between CPS (\$185/hr) and Apex (\$420/hr). Pinnacle is currently the only contractor with mobilization capability at SPP-WIND-025, so the documentation-currency issue cannot be addressed by simple vendor substitution.

3. Root Cause Hypothesis

- **Primary:** Pinnacle's internal procedure-library currency process is broken. The 2022 acquisition created two parallel procedure libraries (NorthStar's and Pinnacle's) that were never merged; technicians default to the library they were trained on.
- **Contributing:** Cascadia's vendor-onboarding process does not require evidence of procedure-library currency review at contract execution. VRA-2023-018 §6 ("Vendor Quality Assurance") is silent on procedure-library audit.
- **Contributing:** Service bulletins are forwarded to Pinnacle but no closeout receipt is required. There is no evidence that NDX-SB-2024-006 was acted on by Pinnacle.

4. Conflict With Existing Guidance

- **NDX-SB-2024-006** (OEM bulletin, binding under VRA-2023-018 §4.2) — not implemented by Pinnacle.
- Pinnacle's referenced procedures (NSW-WPM-04, NSW-WPM-07, NSW-WPM-09, NSW-WPM-12) cite a vendor identity (NorthStar Wind Services) that has not legally existed since the 2022 acquisition.
- **Cascadia Vendor Quality Standard VQS-2024-01 §3.1** requires service contractors to perform work to current-version OEM procedures unless an Engineering deviation is approved. No deviation has been requested by Pinnacle.

5. Recommended Corrective Action

1. Issue formal notice-of-non-conformance to Pinnacle under VRA-2023-018 §6 citing the 11 audited work orders.
2. Require Pinnacle to provide a written procedure-library currency review covering all procedures applicable to Cascadia assets within 90 days. Cascadia Engineering shall review the deliverable.
3. Require Pinnacle to re-perform the missing NDX-SB-2024-006 blade-root bolt torque-verification step on all 26 turbines at SPP-WIND-025 within 60 days, at Pinnacle's expense.

4. Modify VRA-2023-018 at next renewal (2026-Q1) to add procedure-library currency obligations and a service-bulletin closeout-receipt requirement.
5. Establish a standing quarterly Cascadia-side audit (sample 10% of vendor work orders) to verify procedure currency.

6. Action Owner & Timeline

Sub-action	Owner	Target Date
Non-conformance notice to Pinnacle	M. Beasley	2025-11-15 — missed
Procedure-library currency review (vendor deliverable)	Pinnacle (vendor)	2026-01-31
Re-perform NDX-SB-2024-006 on 26 turbines	Pinnacle (at vendor cost)	2025-12-31 — partially complete (14 of 26 as of CAR re-issue)
VRA-2023-018 renewal modification	Legal + Procurement	2026-Q1
Quarterly vendor work-order audit	Reliability & Compliance	2026-Q1 onward

7. Implementation Status

Open. Re-performance of NDX-SB-2024-006 is partially complete; Pinnacle has acknowledged the deficiency in writing (2025-10-15). Procedure-library currency review is in progress. The non-conformance notice has been drafted but not transmitted (held pending the renewal-discussion window). No findings yet on the other 7 NSW-prefixed procedures cited in the audit.

Cross-references: VRA-2023-018 (Pinnacle master agreement); VQS-2024-01 §3.1 (vendor quality standard); NDX-SB-2024-006 (OEM bulletin); CAR-2025-004 (parallel vendor-contract structural finding); CAR-2025-007 (parallel vendor-documentation-currency finding affecting GE); OIR-2025-009 (WIN-10007 / WIN-10132 outage cluster review); RCA-2025-005 (blade-root inspection adequacy review).

End of report.